ARES integration

One contract, then the demonstration. Candidate case study, NEST programme, Open Earth Foundation.

The consortium

Solis
capacity, multi-year horizons

Tharsis
habitat demand, hour by hour

Meridian
failure, thresholds, evidence

Helix
schema translation layer

Francis Li · 29 August 2026

This is triage

Four things have gone wrong at month 8. They are four views of one missing artifact.

An exchange is two months late, in the wrong format.
Cadence, latency and encoding were never contract terms. An exchange cannot be late against a deadline nobody set.

Two partners dispute what variables mean and who owns them.
The variables have no declared owner, so the partner that needed names invented them and built against them.

The risk laboratory says the interface cannot carry its claims.
It has no uncertainty clause. A missing clause, not a complaint about quality.

Nobody has shown the coupling is necessary.
The operators bridging hourly simulation to multi-year planning are owned by no one.

Second-year funding depends on a demonstration. This is a crisis to stabilise, not a programme to improve.

Four objects nobody owns

Read off the partners' own consumes and produces clauses. Nothing inferred.

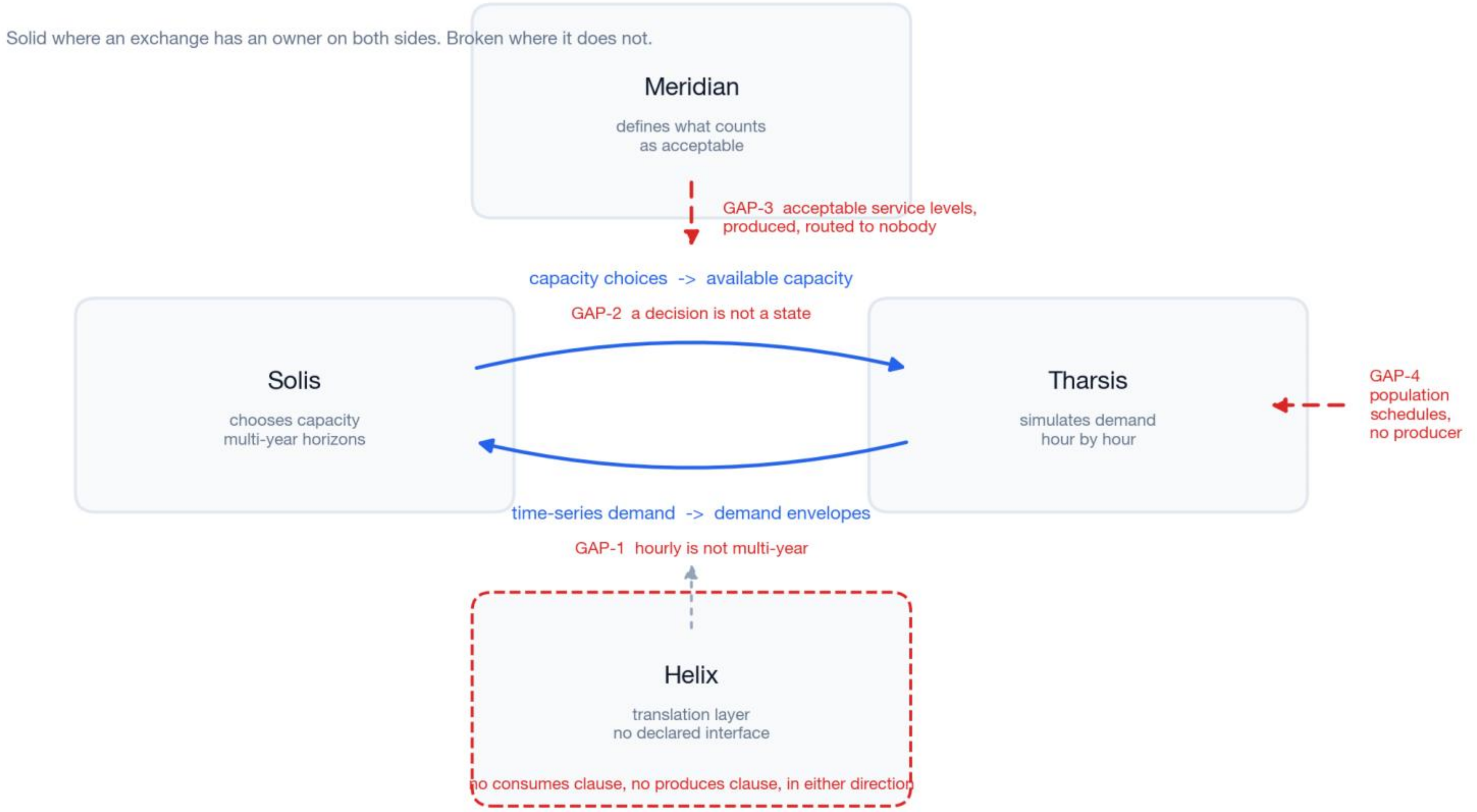
GAP-1.
Hourly trace to multi-year bound. Nobody owns the conversion.

GAP-2.
Installed capacity to available capacity. Tharsis derates privately.

GAP-3.
Thresholds and stress scenarios produced, routed to nobody.

GAP-4.
The largest demand driver, produced by no partner.

GAP-5.
Which loads can be shed under stress, and which cannot.



At what fidelity does this become admissible evidence?

Not whether tight coupling is interesting. Where the line sits between a model that can carry a decision that kills people if it is wrong, and one that cannot.

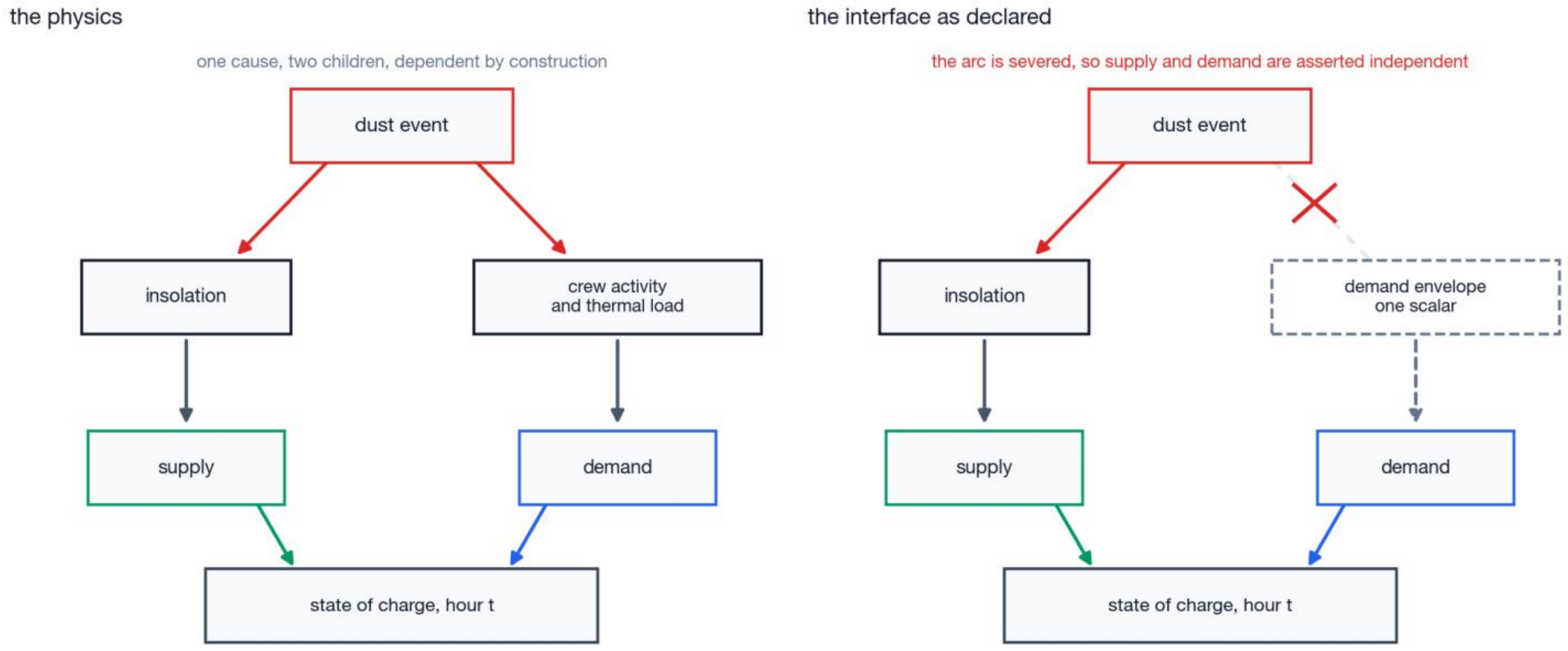
A settlement model exists so somebody can sign a habitat design.
Below a level of resolution and coupling it is not a coarser instrument. It is inadmissible as evidence.

No engineer signs a life-safety design on a tool its own authors would not trust.
That sets a floor, and this consortium is below it.

The physics forces the same answer.
A dust event cuts generation and drives crew indoors at once. Supply and demand answer to one cause.

Exchange bounds once and that dependence is severed by construction.

Meridian's clause 4.2 already forbids it: where two stressors share a physical cause, the model shall not treat them as independent.



Meridian, clause 4.2: where two stressors share a physical cause, the model shall not treat them as independent.

Six weeks, and never nothing to show

From week 2 the demonstration runs. Every week after improves it rather than building toward a first one.

Week 2 runs end to end on crude inputs.
Placeholder numbers, one sol, nothing polished. Its job is to make every format and calendar mismatch fail while there is time.

The consortium is at month 8 having never run end to end. A late exchange in an unexpected format is what that produces.

Each week's output is a superset of the last.
The worst case in week 6 is showing week 5's version.

Week 5 is buffer.
Nothing is scheduled into it that cannot move out.

Week 5 also tests the recommendation.
If Solis's revision comes back inside a tolerance set in week 1, the feedback is weak and the coupling argument is wrong.

Two tracks, and where the slack is

Solis to Tharsis is the critical path. Meridian runs alongside it. Week 5 absorbs a slip and week 6 rehearses.

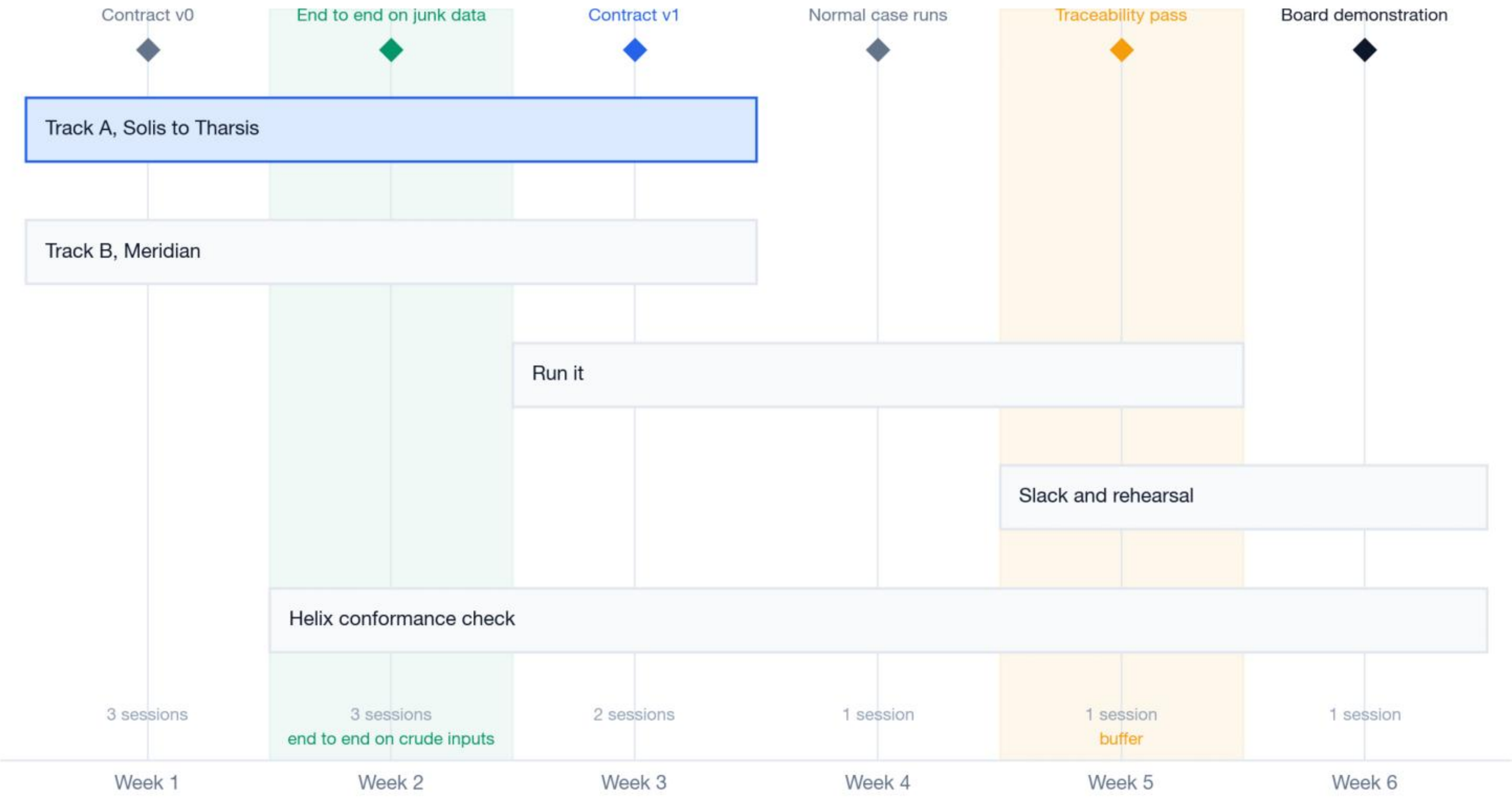
Track A
is the critical path. The exchange that already failed.

Track B
runs alongside, because what Meridian requires changes what track A must carry.

Weeks 3 and 4.
Normal case, then the stress event run iterated.

Week 6.
The traceability pass. For every number: which model, which contract, which schedule version, which threshold set.

Every action closes a named gap, so the plan is traceable to the diagnosis.



Who is in the room, and who decides

Eleven sessions in six weeks. Being present and holding the ruling are different things, and the difference is the governance.

Weeks 1 and 2 carry five of the eleven
including the derate session, which needs Solis and Meridian in one room and is the expensive decision in the plan.

Two to three touchpoints a week
then a weekly thirty minute standup through week 5.

Helix is in seven of eleven rooms and holds none of the rulings.
It must be present when the contract it built against is settled. It cannot hold the decision.

It also holds the only working transport code for the exchange that failed. We cannot tell from here whether that helps, so the plan gains if it does and loses nothing if it does not.

Filled where the partner is in the room. A dot marks who holds the ruling. Shaded columns settle what a variable means or who owns it.

Nobody has to be persuaded

Every partner is worse off under the status quo, and three do not yet know how.

Helix
built against a provisional reading. A declared contract is the only outcome where that work survives. Approach first.

Meridian
has already asked for the uncertainty clause, in the form of an objection. Agree with it in public.

Solis
is judged on a derate function it never saw. Declaring its units is how it takes back control of how its own design is evaluated.

Tharsis
carries blame for thresholds it was never sent and a schedule nobody owns. Declaring the routes moves that liability where it belongs.

In an acute crisis partners accept direction they would decline in business as usual, and that window closes when the crisis is declared over. Week 1 uses it.

What I built, and what it cannot tell you

430 clause cells across 43 declarations. 35 are stated.

It reads the partners' own documents.
Four declarations in four incompatible formats, then seven general rules. No rule names a partner, an item or a gap, so it was never told what to find.

Five clauses are absent in all 43 declarations
uncertainty, spatial resolution, scenario identity, cadence, encoding.

The consortium yields one identifiable exchange, and it is broken three ways at once.

Deterministic rule code, no model inside it.
A finding four disagreeing institutions must accept has to be reproducible byte for byte.

What it cannot tell you.
Which variables are actually disputed. Nothing in the record says, and that ARES cannot answer it in month 8 is itself the finding.